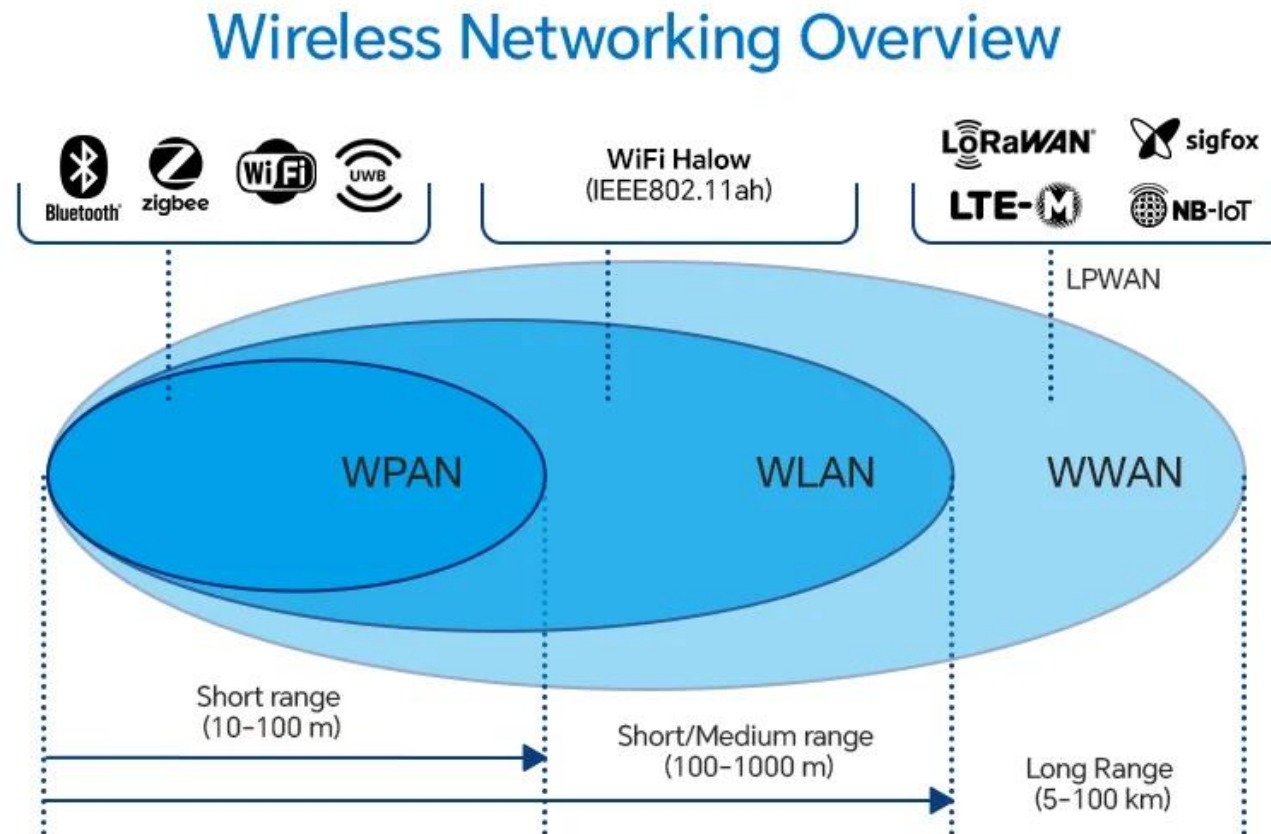


Introduction to Internet of Things

Lecture 18&19

WPAN Technologies for IoT

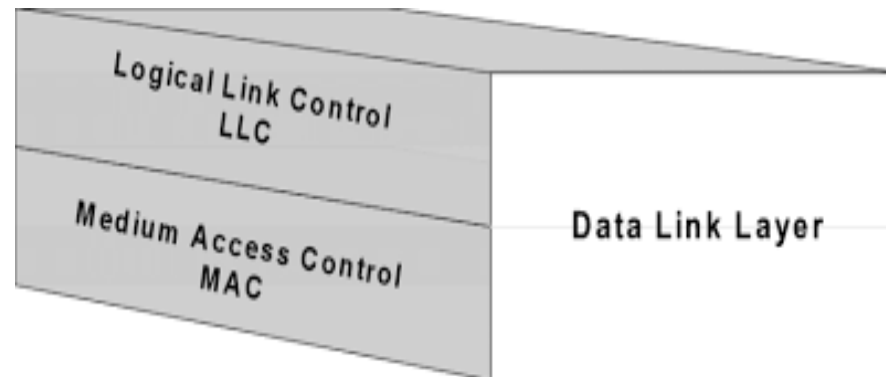
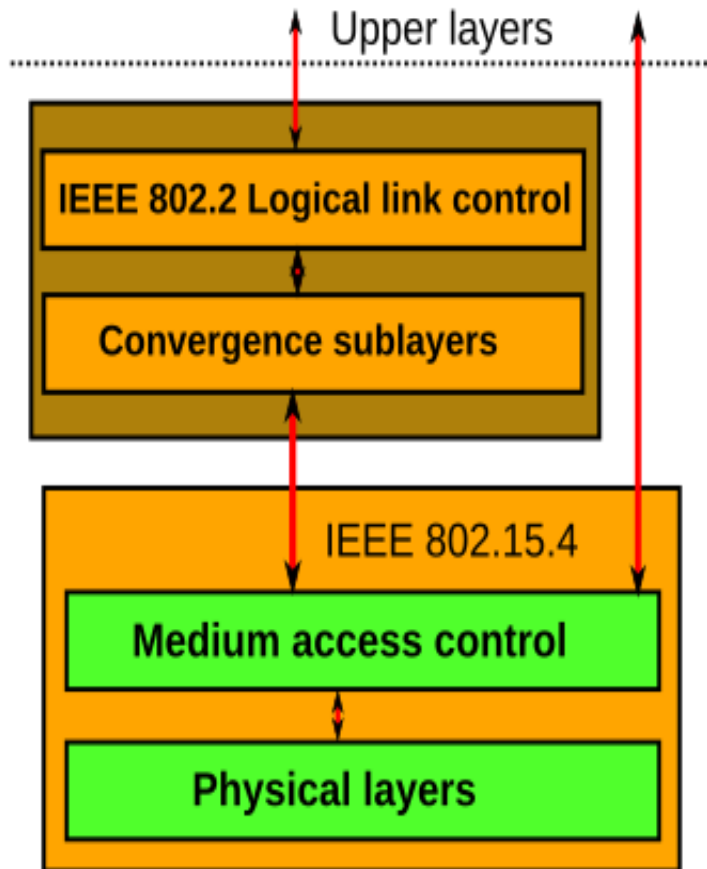
- ▶ Short range
- ▶ Low power
- ▶ Low cost



IEEE 802.15.4 (LR-WPAN)

Features:

- ▶ Its ability to support many devices in a network--- star topology
- ▶ Its support for low power operation.
- IEEE 802.15.4 is a popular choice for WPAN technology in IoT due to its low power consumption, reliability, and ability to support many devices.
- The IEEE 802.15.4 standard defines the physical and MAC layer communication methods used to access wireless networks.



IEEE 802.15 WPAN

► MAC (Media Access Control)

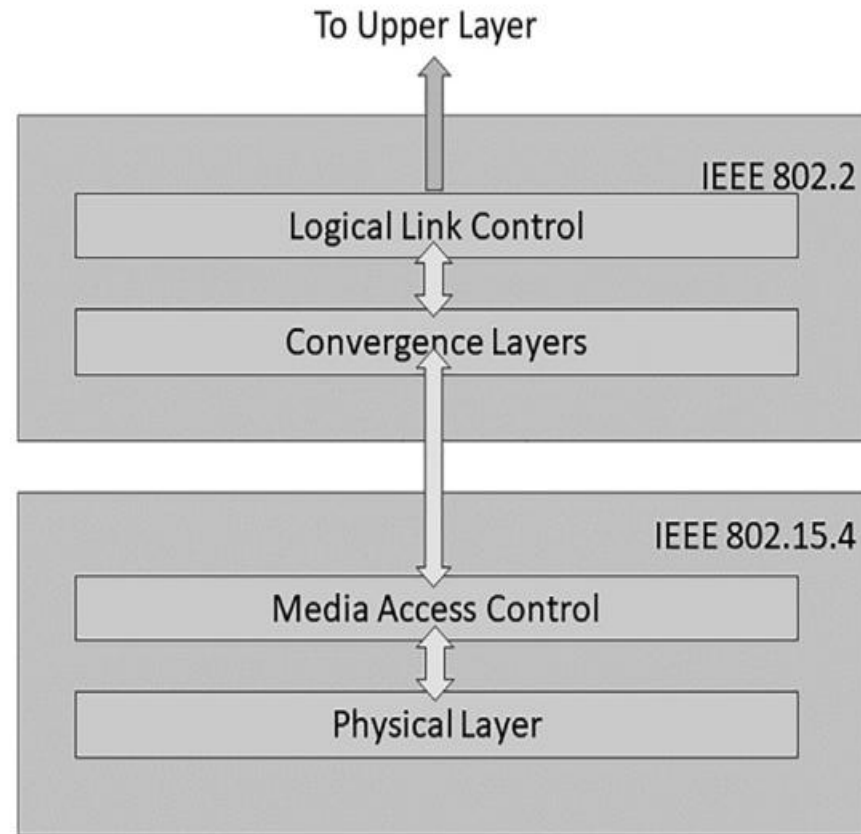
Controls how devices access the medium

- Frame transmission
- MAC addresses
- Channel access (CSMA/CA, etc.)

► LLC (Logical Link Control)

Provides interface to upper layers

- Flow control
- Error control
- Multiplexing protocols



IEEE 802.15.4

- ▶ This is a specific standard inside 802.15
- ▶ Designed for:
 - Low power
 - Low data rate
 - IoT / sensor networks
- ▶ It defines PHY + MAC layers for low-power devices

MAC layer	
802.11	WiFi
802.15.1	Bluetooth
802.15.4	ZigBee, SLowPAN
802.16	Wireless Metropolitan Access Networks Broadband Wireless Access (BWA) WiMax

MAC functions

1. Beacon Management
2. Channel Access
3. Frame Formatting
4. Acknowledgments
5. Network Topology

Key Characteristics of IEEE 802.15.4

- IEEE 802.15.4, a wireless communication standard, exhibits several key characteristics that make it particularly suited for specific applications. Here are the essential characteristics of IEEE 802.15.4:
- **1. Low Data Rates:**
- IEEE 802.15.4 is optimized for applications with low data rate requirements. It typically supports data rates in the range of a few kilobits per second (Kbps) to a few hundred Kbps.
- This characteristic is ideal for transmitting small packets of data efficiently, making it suitable for applications like sensor data collection and monitoring.
- **2. Low Power Consumption:**
- Energy efficiency is a hallmark of IEEE 802.15.4. Devices operating under this standard are designed to minimize power consumption.
- Devices can spend a significant portion of their time in low-power sleep modes and wake up only when necessary, extending battery life. This is especially valuable for battery-powered devices and energy-harvesting solutions.
- **3. Short-Range Communication:**
- IEEE 802.15.4 is intended for relatively short-range wireless communication, typically within a range of 10 to 100 meters.
- Short-range communication is well-suited for applications where devices are in close proximity, such as within a room or a small area.
- These key characteristics collectively make IEEE 802.15.4 an excellent choice for applications where conserving power, achieving low data rates, and maintaining short-range communication are essential. Examples of such applications include wireless sensor networks, industrial automation, home automation (smart home devices), and various IoT deployments. IEEE 802.15.4's efficiency in addressing these specific requirements contributes to its widespread adoption in these domains.

Network Topologies in IEEE 802.15.4

- IEEE 802.15.4, a versatile wireless communication standard, supports various network topologies to accommodate a wide range of application requirements. Here, we'll explore the primary network topologies supported by IEEE 802.15.4:
- **1. Star Topology:**
- **Description:** In a star topology, all devices communicate directly with a central hub or coordinator. The central hub acts as a coordinator and manages communication between devices.
- **Advantages:**
 - Simplicity: Easy to set up and manage, making it suitable for simple, point-to-point applications.
 - Centralized control: The coordinator can efficiently manage communication, making it suitable for applications with low latency requirements.
- **Use Cases:** Home automation, sensor networks in small areas, smart lighting systems.

Network Topologies in IEEE 802.15.4 (

- **2. Peer-to-Peer (P2P) Topology:**
- **Description:** In a peer-to-peer topology, devices can communicate directly with each other without the need for a central coordinator. Devices act as equals and can send data directly to one another.
- **Advantages:**
 - Decentralized: No single point of failure, making it robust and suitable for applications requiring redundancy.
 - Low latency: Devices can communicate directly, reducing communication delay.
- **Use Cases:** Device-to-device communication, ad-hoc networks, short-range data transfer.
- **3. Mesh Topology:**
- **Description:** A mesh topology is characterized by devices interconnected with multiple paths. Devices in a mesh network can relay data for each other, creating a self-healing and robust network.
- **Advantages:**
 - Reliability: Redundant paths ensure data can reach its destination even if one path fails, improving network reliability.
 - Extended range: Mesh networks can cover larger areas than star or P2P networks due to data relay.
- **Use Cases:** Wireless sensor networks, industrial automation, smart city applications.

Network Topologies in IEEE 802.15.4

- **4. Cluster Tree Topology:**
- **Description:** A cluster tree topology combines elements of both star and mesh topologies. It features a central coordinator (the root) that communicates with multiple cluster heads. Each cluster head, in turn, communicates with leaf nodes.
- **Advantages:**
 - Scalability: Suitable for large networks with hierarchical structures.
 - Efficient use of power: The coordinator and cluster heads can efficiently manage network traffic and power consumption.
- **Use Cases:** Zigbee-based smart home networks, industrial control systems.
- **5. Hybrid Topologies:**
- **Description:** Some IEEE 802.15.4 networks use hybrid topologies, combining multiple network structures to meet specific application needs.
- **Advantages:**
 - Flexibility: Provides the flexibility to adapt to complex network requirements.
 - Customization: Allows network designers to optimize performance for their specific use cases.
- **Use Cases:** Large-scale IoT deployments, complex industrial automation systems.

Applications of IEEE 802.15.4

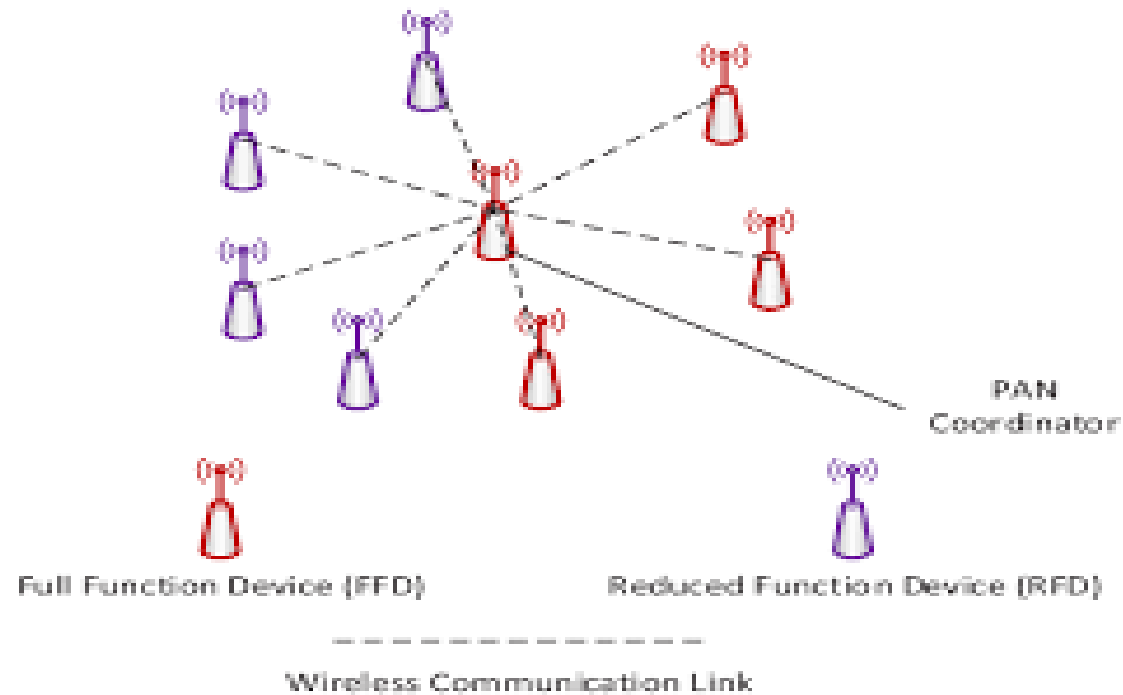
- IEEE 802.15.4, known for its low-power and short-range wireless capabilities, is employed in a variety of applications across different industries. Here are some common applications of IEEE 802.15.4:
- **1. Wireless Sensor Networks (WSNs):**
- **Description:** WSNs consist of sensor nodes equipped with IEEE 802.15.4 transceivers that collect data from the physical world and transmit it wirelessly to a central coordinator or sink node.
- **Use Cases:** Environmental monitoring (e.g., temperature, humidity, air quality), precision agriculture, wildlife tracking, structural health monitoring.
- **2. Industrial Automation:**
- **Description:** IEEE 802.15.4 is extensively used in industrial settings to connect various sensors, controllers, and actuators. It enables real-time data collection and control in manufacturing and process automation.
- **Use Cases:** Factory automation, asset tracking, predictive maintenance, remote monitoring of industrial equipment.

Applications of IEEE 802.15.4

- **3. Home Automation (Smart Home):**
- **Description:** In smart homes, IEEE 802.15.4 enables wireless communication between smart devices, enhancing convenience, energy efficiency, and security.
- **Use Cases:** Smart lighting systems, smart thermostats, door locks, security cameras, and home energy management.
- **4. Healthcare and Medical Monitoring:**
- **Description:** IEEE 802.15.4 is employed in wearable and implantable medical devices to monitor patients' vital signs and transmit data to healthcare providers or caregivers.
- **Use Cases:** Remote patient monitoring, fall detection, medication adherence tracking.
- **5. Environmental Monitoring:**
- **Description:** Environmental monitoring applications rely on IEEE 802.15.4 to collect data from sensors placed in remote or hazardous locations.
- **Use Cases:** Forest fire detection, air quality monitoring, water quality monitoring in rivers and lakes.

802.15.4

- 802.15.4 uses carrier-sense multiple access with collision avoidance (CSMA/CA).



Nodes of 802.15.4

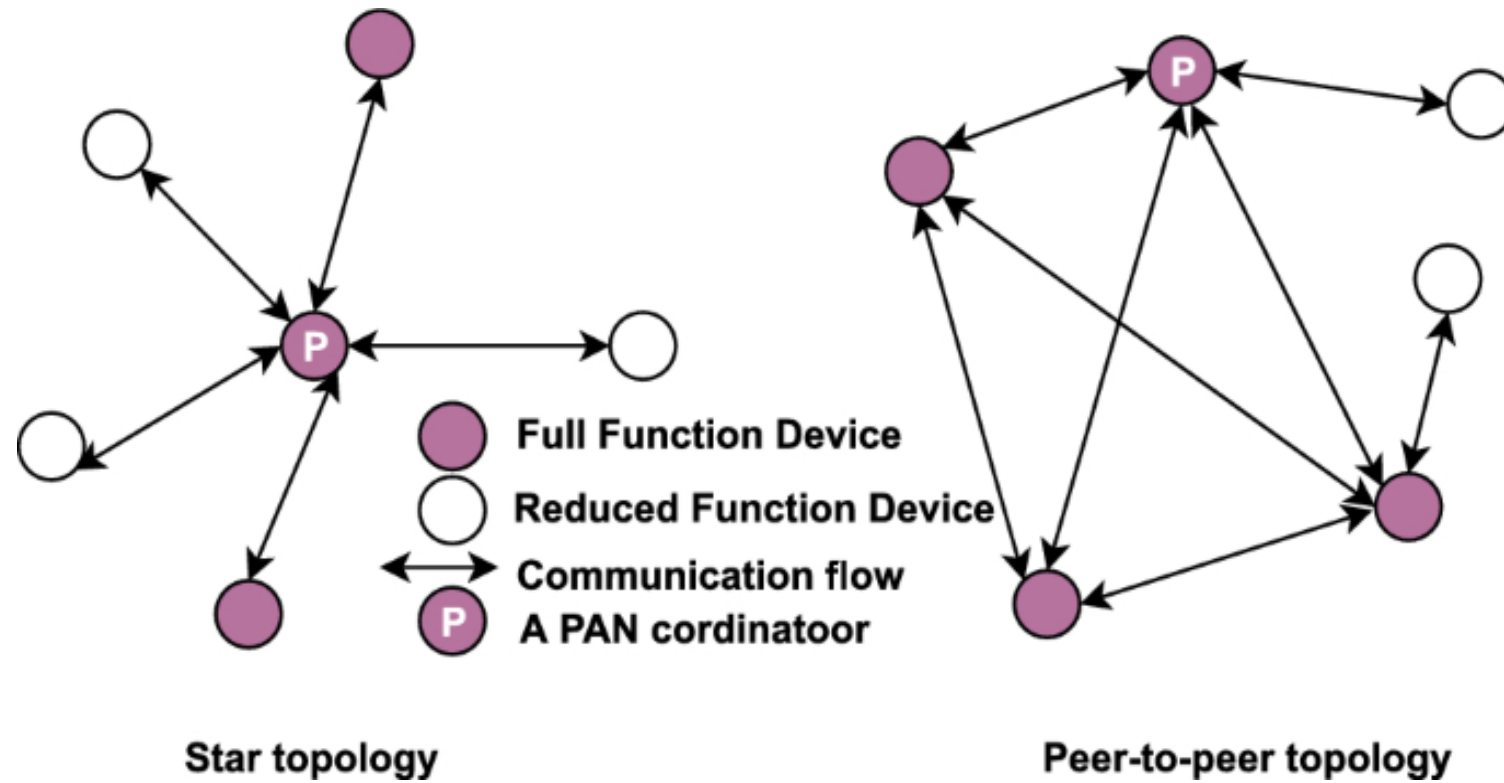
- ▶ **PAN coordinator:**
- ▶ 802.15.4 networks are setup by a *PAN coordinator* node/coordinator.
- ▶ There is a single PAN coordinator for each network identified by its PAN ID.
- ▶ The PAN coordinator is responsible for scanning the network and selecting the optimal RF channel, and for selecting the 16 bits PAN ID (personal area network identifier) for the network.
- ▶ Other 802.15.4 nodes must send an association request for this PAN ID to the PAN coordinator in order to become part of the network.

- ▶ **FFD (Full Functional devices):**
- ▶ These devices are capable of relaying messages to other FFDs, including the PAN coordinator.
- ▶ The first coordinator to send a beacon frame becomes the PAN coordinator.
- ▶ Other devices join the PAN coordinator as their parent, and among those devices the FFDs also begin to transmit a periodic beacon, or to respond to beacon requests.
- ▶ At this stage more devices may be able to join the network, using the PAN coordinator or any FFD as their parent.

- ▶ ***Reduced Function Devices (RFD):***
- ▶ They cannot route messages.
- ▶ Usually, their receivers are switched off except during transmission.
- ▶ They can be attached to the network only as leaf nodes.

Each network, identified by its PAN ID, is called a ***cluster***

Two topologies in 802.15.4



Access control methods

- ▶ The MAC layer specified by 802.15.4 defines two access control methods for the network.
- ▶ The ***beacon-enabled access method (or slotted CSMA/CA)***. When this mode is selected, the PAN coordinator periodically broadcasts a *superframe*, composed of a starting and ending beacon frame, 15 time slots, and an optional inactive period during which the coordinator may enter a low-power mode.
- ▶ The ***nonbeacon-enabled access method (unslotted CSMA/CA)***. This is the mode used by ZigBee and 6LoWPAN. All nodes access the network using CSMA/CA. The coordinator provides a beacon only when requested by a node.

Uses of 802.15.4

- ▶ 802.15.4 provides all the MAC and PHY level mechanisms required by higher level protocols to exchange packets securely.
- ▶ It does not provide a fragmentation and reassembly mechanism.
- ▶ Bandwidth is also very limited, Packets cannot be sent continuously.
- ▶ With these limitations in mind, 802.15.4 is clearly targeted at sensor and automation applications.

Zigbee

- ▶ Zigbee, commonly used in IoT applications due to its low-power, low-data-rate capabilities, and ability to support large-scale mesh networks.
- ▶ It utilizes low-duty cycles and sleep modes, allowing devices to conserve energy and operate on battery power for extended periods.
- ▶ In comparison, protocols like Wi-Fi and Bluetooth are more power-hungry, making them less suitable for battery-operated IoT devices.
- ▶ Zigbee is primarily used for two-way communication between a sensor and a control system.

Features of Zigbee

- ▶ Low Power Consumption
- ▶ Mesh Networking
- ▶ Large-Scale Deployments
- ▶ Security
- ▶ Interoperability

Zigbee resides on top of the PHY and MAC layers defined in IEEE 802.15.4.

Application Layer	ZigBee
Network Layer (Routing/ Network)	ZigBee
Data Link Layer (MAC – CSMA/CA)	IEEE 802.15.4
Physical Layer (2.4GHz, 250kbps)	IEEE 802.15.4

▪ ZigBee supports different network topologies like:

- STAR
- Peer to Peer (Mesh)
- Cluster Tree

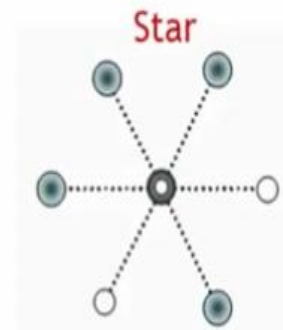
Layers

1. Network and Application Support layer
2. Physical (PHY) layer
3. Media access control (MAC) layer

Network Topologies

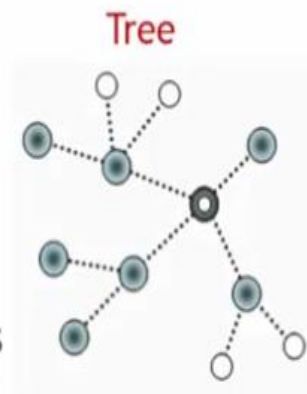
▪ Star

- Network controlled and initiated by central PAN (Personal Area Network)
- End Devices communicate directly with Coordinator



▪ Tree

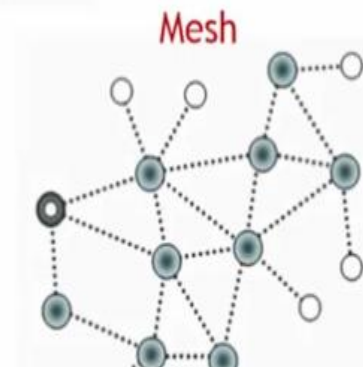
- Tree structure rooted at the PAN coordinator
- Coordinator initiates network
- Children route through Parents in strict hierarchy
- Not always appropriate for low-latency applications



- Coordinator
- Full Function Device (FFD)
- Reduced Function Device (RFD)

▪ Mesh

- Messages may be routed from any source to any destination
- Every fixed function device can be a router for its neighbors
- Multi-hop topology extends range of the network
- High reliability achieved through multiple paths



Architecture

- Layered architecture
- These layers facilitate the features that make ZigBee very attractive:
 - ✓ low cost
 - ✓ easy implementation
 - ✓ reliable data transfer
 - ✓ short-range operations
 - ✓ very low power consumption
 - ✓ adequate security features

Zigbee Device types

- There are three different ZigBee device types that operate on the layers in any self-organizing application network

1. Zigbee Coordinator node
2. Full Function Device (FFD)
3. Reduced Function Device(RFD)

Zigbee coordinator node

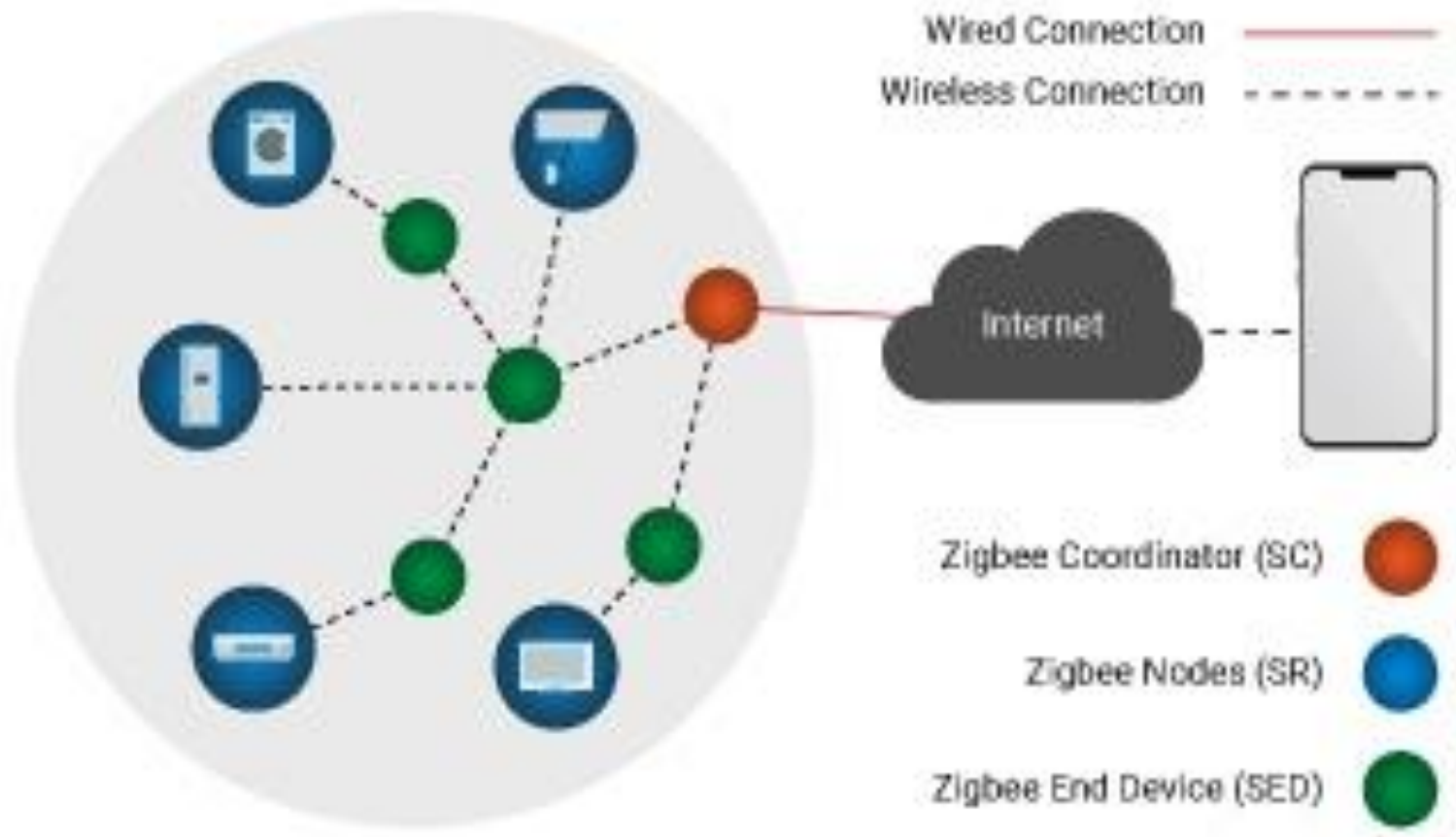
- It is the root of network tree and a bridge to other network
- Able to store information about the network
- Only one ZCN for a network
- It act as a repository for other security keys

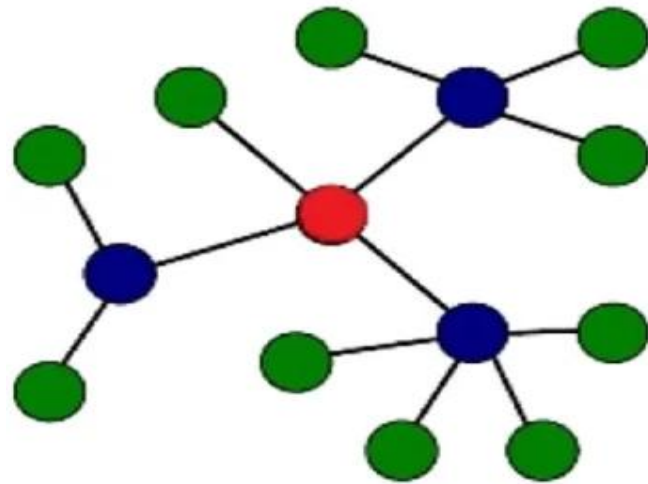
Zigbee full functional device

- An intermediary router transmitting data from other devices
- Needs lesser memory than Zigbee coordinator node
- Lesser manufacturing cost
- Can operate on all topologies
- Also act as a coordinator

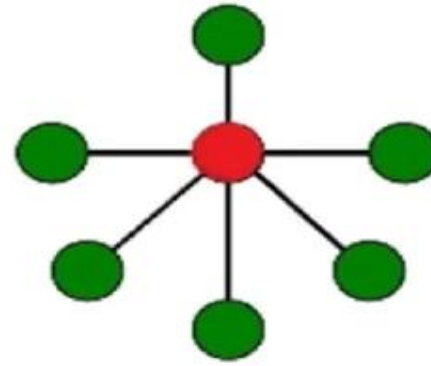
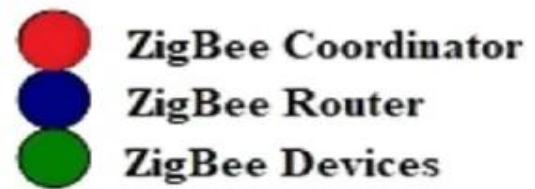
Zigbee Reduced function device

- Device capable of talking in the network
- It cannot relay data from other devices
- Less memory
- Cheaper than FFD
- It talks only to the n/w coordinator

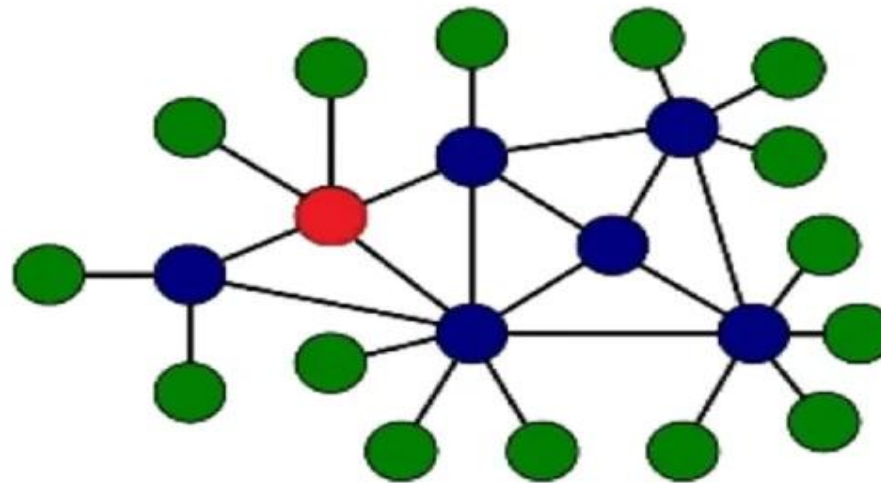




Cluster Tree Network



Star Network



Mesh Network

Characteristics

- Low power consumption with battery life ranging from months to years
- High density of nodes per network
- Low cost
- Simple implementation
- Low data rate
- Small packet devices